

Model-assisted estimation under the Monte Carlo approach

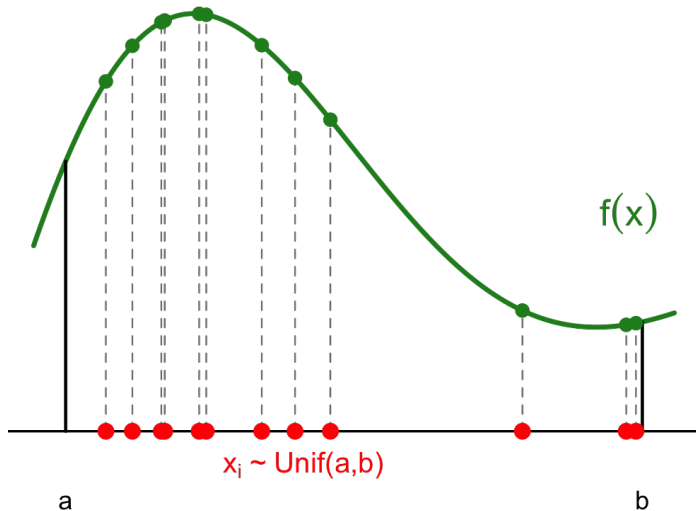
From the external model assumption to the classical regression estimator

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Monte Carlo integration basics



Steps

- 1 Draw $x_1, \dots, x_n$ uniform & independent on $[a, b]$
- 2 Evaluate $f(x_i)$
- 3 Average: $\bar{f} = \frac{1}{n} \sum_i f(x_i)$
- 4 Scale by domain length:

$$\int_a^b f(x) dx \approx (b-a) \bar{f}$$

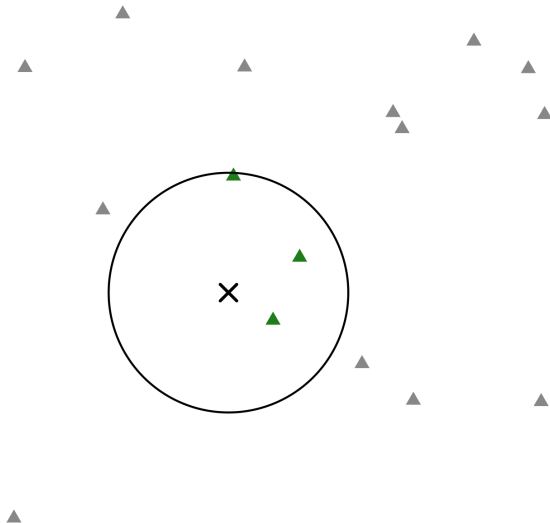
Law of Large Numbers

$$\bar{f} \rightarrow \mathbb{E}[f(x)]$$

Central Limit Theorem

$$\sqrt{n}(\bar{f} - \mathbb{E}[f(x)]) \xrightarrow{d} \mathcal{N}(0, \sigma^2) \text{ (for CLs)}$$

Sample unit is a point



- A circular plot is centered on the **sample point** selects trees inside the inclusion radius.
- Multiple inclusion radii are possible based on DBH thresholds.
- There are infinite points in the forest.
- **Finite population sampling theory** is commonly used but the problem is poorly posed.
- Circles do not tessellate! Angle count sampling makes no sense...

Workaround: Draw same inclusion rings around trees

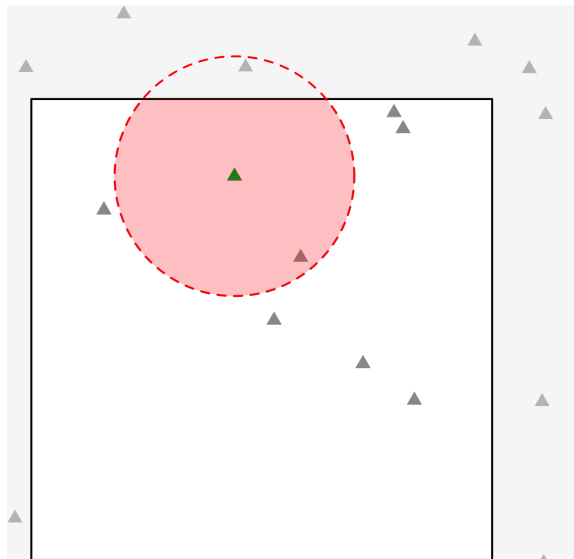
Definition (Duality principle)

trees in radius around point same as point in radius around trees.

- Overlap of the inclusion rings illustrates density accumulation at a point.
- Sample unit is the point; not the trees.

x point at plot center

Boundary adjustment on trees (not plots)



- Plots with centers outside forest (gray) are **not visited** for budget reasons.
- Requires boundary adjustment for **trees**.

Definition (Indicator variable)

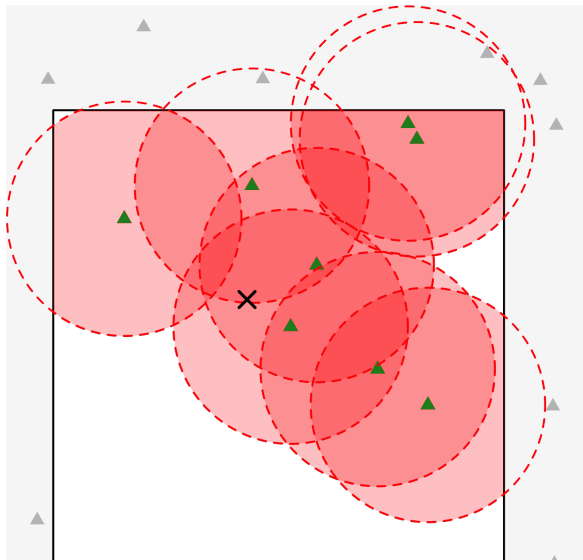
$$I_i(x) = \begin{cases} 1, & \text{if } x \in K_i, \\ 0, & \text{if } x \notin K_i. \end{cases}$$

Definition (Inclusion probability)

$$\pi_i = \mathbb{P}[I_i(x)] = \mathbb{E}[I_i(x)] = \frac{A_{K_i}}{A_T}$$

K_i inclusion ring around tree i
 A_T ; A_{K_i} area of T ; K_i (bnd. adjusted)
 T territory of interest

Local density surface from above



Surface is fixed; point is random

Definition (Local density surface)

$$Y(x) = \frac{1}{A_T} \sum_{i=1}^N \frac{I_i(x) Y_i}{\pi_i}$$

Y_i measurement for tree i

π_i inclusion probability for tree i

N number of trees in territory T

$I_i(x)$ indicator that tree i is selected at x

A_T surface area of T

Interpretation

$Y(x)$ is the Horvitz–Thompson estimator (for the spatial) mean with sample size one.

Exercise — Prove equivalence of sampling trees and Monte Carlo integration

Show (assume uniform independent sampling)

$$A_T \cdot \mathbb{E}[Y(x)] = \int_T Y(x) dx = \sum_{i=1}^N Y_i$$

Hint : $\mathbb{E}[g(X)] = \int g(x) p(x) dx$ and $\mathbb{E}[I_i(x)] = \pi_i$

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Hint : $\mathbb{E}[g(X)] = \int g(x) p(x) dx$ and $\mathbb{E}[I_i(x)] = \pi_i$

Proof.

$$A_T \cdot \mathbb{E}[Y(x)] = A_T \cdot \int_T Y(x) \underbrace{\frac{1}{A_T}}_{p(x)} dx = \int_T Y(x) dx \quad (1)$$

(2)



Exercise — Prove equivalence of sampling trees and Monte Carlo integration

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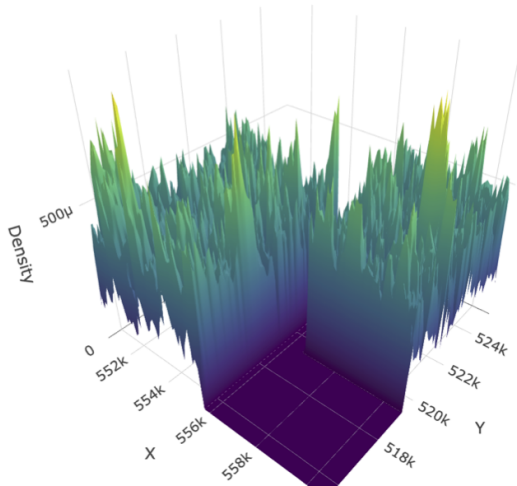
$$A_T \cdot \mathbb{E}[Y(x)] = A_T \cdot \int_T Y(x) \underbrace{\frac{1}{A_T}}_{p(x)} dx = \int_T Y(x) dx \quad (1)$$

$$A_T \cdot \mathbb{E}[Y(x)] = A_T \cdot \mathbb{E} \left[\frac{1}{A_T} \sum_{i=1}^N \frac{I_i(x) Y_i}{\pi_i} \right] = \sum_{i=1}^N \frac{Y_i}{\pi_i} \mathbb{E}[I_i(x)] = \sum_{i=1}^N \frac{Y_i}{\pi_i} \cdot \pi_i = \sum_{i=1}^N Y_i \quad (2)$$



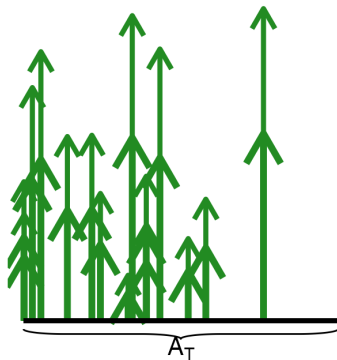
View of 3d local density surface from artificial forest

Not so useful... It is easier to simplify the dimension.

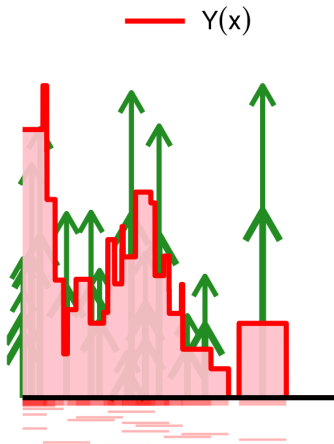


Heuristic representation of Monte Carlo approach

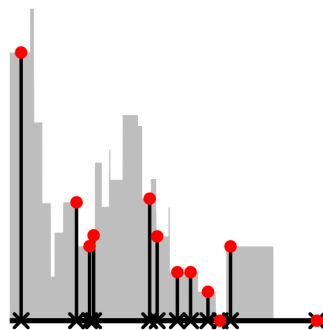
Trees in T



Local Density Surface

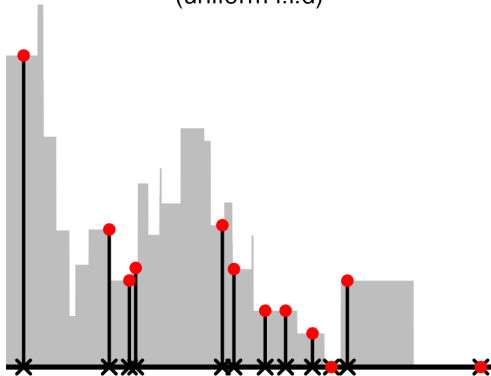


Monte Carlo Integration
(uniform i.i.d)

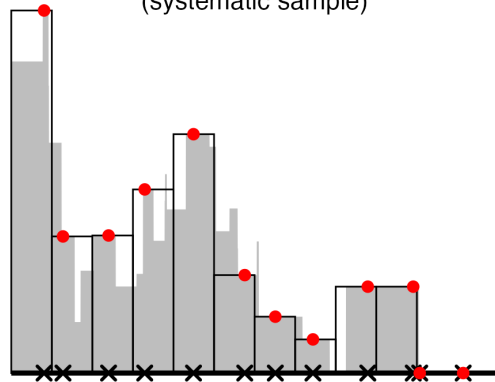


Heuristic representation of Monte Carlo approach

Monte Carlo Integration
(uniform i.i.d)



Riemann Integration*
(systematic sample)



*no design-unbiased variance estimator, but better convergence

One-phase estimation for uniform independent sampling

Population parameters and their sample-copy estimators

Assume A_T is known.

Population	$\mathbb{E}[Y(x)]$	$=$	$\frac{1}{A_T} \int_T Y(x) dx$	$\mathbb{V}[Y(x)]$	$=$	$\frac{1}{A_T} \int_T (Y(x) - \mathbb{E}[Y(x)])^2 dx$
"Sample copy"	$\bar{Y}_s$	$=$	$\frac{1}{n} \sum_{x \in s} Y(x)$	$\hat{\mathbb{V}}_s[Y(x)]$	$=$	$\frac{1}{n-1} \sum_{x \in s} (Y(x) - \bar{Y}_s)^2$

One-phase estimation for uniform independent sampling

Population parameters and their sample-copy estimators

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Definition (One-phase point and variance estimator)

Draw $x \in s$, n times, uniformly and independently.

$$\hat{Y}_{1p} = \frac{1}{n} \sum_{x \in s} Y(x) \quad \mathbb{V}[\hat{Y}_{1p}] = \frac{\mathbb{V}[Y(x)]}{n} \quad \hat{\mathbb{V}}[\hat{Y}_{1p}] = \frac{\hat{\mathbb{V}}_s[Y(x)]}{n}$$

Every point estimator has a population variance estimated by a (sample) variance estimator.

Pros and cons of one-phase estimation

Pros

- **Simple and robust:** the sample mean is consistent by the LLN
- **Few assumptions:** only sampling design and error-free measurements are required
- **Straightforward inference:** confidence intervals follow from the CLT

$$\hat{Y}_{1p} \pm z_{1-\alpha/2} \sqrt{\frac{\hat{V}_s[Y(x)]}{n}}$$

Cons**

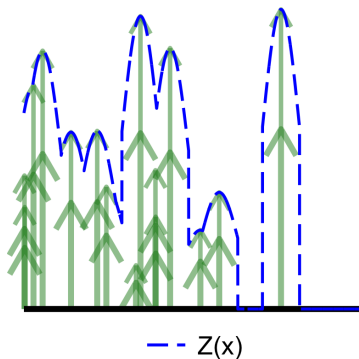
- **Asymptotic:** good approximation requires a sufficiently large (n)
- **Inefficient:** ignores information beyond the sampled ($Y(x)$)

How can we do better ?

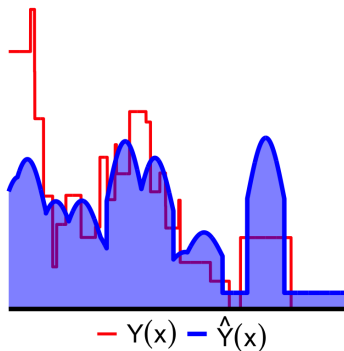
Improve precision with known auxiliary information ("*Model-assisted estimation*").

Model-assisted estimation under the Monte Carlo approach

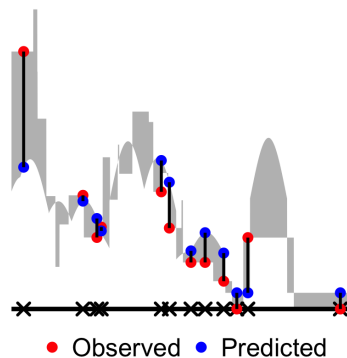
Auxiliary Variable
(e.g. CHM)



Predicted density surface
("known" integral)



MC integration on residual
 $R(x) = Y(x) - \hat{Y}(x)$



Better precision when $\mathbb{V}[R(x)] < \mathbb{V}[Y(x)]$

The difference estimator

Sampling design

Uniform iid sampling

Assisting model

$$Y(x) := m(x) + R(x)$$

$$\hat{Y}(x) := m(x) \quad (\text{any model fit indep. of } s)$$

Point estimator

$$\hat{Y}_{diff} = \frac{1}{A_T} \int_T \hat{Y}(x) dx + \frac{1}{n} \sum_{x \in s} R(x)$$

Population variance

$$\begin{aligned} \mathbb{V}[\hat{Y}_{diff}] &= \mathbb{V} \left[\frac{1}{A_T} \int_T \hat{Y}(x) dx + \frac{1}{n} \sum_{x \in s} R(x) \right] \\ &= \frac{1}{n} \mathbb{V}[R(x)] \end{aligned}$$

Variance estimator

$$\hat{\mathbb{V}}[\hat{Y}_{diff}] = \frac{1}{n} \hat{\mathbb{V}}_s[R(x)]$$

Example: “Model-assisted” using plug-in estimation:

Consider a linear model :

$$Y(x) := \mathbf{Z}(x)^t \beta + R(x) \quad \text{"assisting model" proposal}$$

$$\hat{Y}(x) = \mathbf{Z}(x)^t \hat{\beta} \quad \hat{R}(x) = Y(x) - \hat{Y}(x) \quad \text{Plug-in to } \hat{Y}_{diff}$$

- $\hat{\beta}$ fit independently of s (i.e. external) $\implies$ exactly design-unbiased.
- $\hat{\beta}_s$ fit includes s (i.e. internal) $\implies$ asy. design-unbiased (for "most" choices of $\hat{\beta}_s$).

Note: better variance estimators likely exist (g-weight, bootstrap, etc)

Definition (External model)

An assisting model $\hat{Y}(x) = \hat{m}(x)$ is **external** if the training set used to fit it is independent of the sample s in target territory T .

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An assisting model $\hat{Y}(x) = \hat{m}(x)$ is **external** if the training set used to fit it is independent of the sample s in target territory T .

- If the model is external, the $\hat{Y}_{diff}$ and variance estimator are **exactly design-unbiased**.
- This holds for **any** model class $\hat{m}(\cdot)$ — parametric or not:
 - Linear regression
 - k NN
 - CART / random forests
 - A cat with the laser pointer
 - ... and anything else you can fit externally.
- The model can be **wrong** because it simply shifts the Monte Carlo integration to the residuals.
- Precision increases if $\mathbb{V}[Y(x)] > \mathbb{V}[R(x)]$.

Close your eyes and treat internal as external...

The "external model assumption"

If an **internal** model (fit using s itself, or data dependent on it) is *treated as if* it were external, we are making the **external model assumption**.

- Only **asymptotically** design-unbiased results for most reasonable model classes.
- Asymptotic validity is likely but not automatic — **someone has to do the work** to demonstrate it holds for the specific model class being used.
- See Breidt & Opsomer, 2017 for demonstration for many modern modeling classes.

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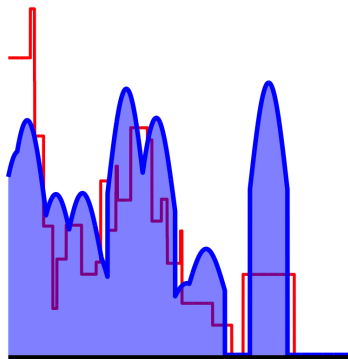
Takeaway

External $\implies$ exact design-unbiasedness, no restrictions on the model.

Internal, assumed external $\implies$ asymptotic unbiasedness when justified for model-class.

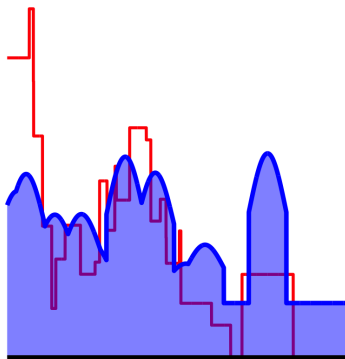
Internal models are not fixed under different sampling realizations

Seed 1



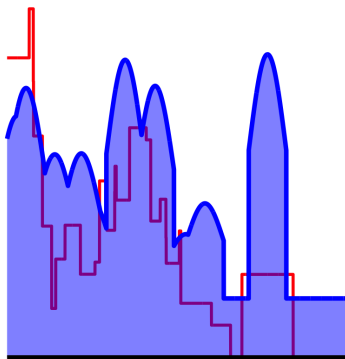
$$\text{— } \hat{Y}(x) = Z(x)^t \hat{\beta}$$

Seed 2



$$\text{— } \hat{Y}(x) = Z(x)^t \hat{\beta}$$

Seed 3



$$\text{— } \hat{Y}(x) = Z(x)^t \hat{\beta}$$

How do we account for the internal-fit contribution to variance?

Two common approaches:

Bootstrap

Refit model in each bootstrap sample.

- SRS with replacement
- Repeat entire modeling procedure
- Very simple and generalizable
- Cumbersome for many territories and/or parameters

Specify a model class

e.g. linear regression

- Taylor approximation for $\hat{\beta}_s$ around β
- This is called the **g-weight** variance
- Good statistical properties (i.e. *calibration*)

Design-based covariance of $\hat{\beta}$

Under uniform i.i.d. sampling, consider a **linear model**,

$$Y(x) = \mathbf{Z}(x)^\top \beta + R(x) \qquad \hat{\beta}_s = \left\{ \frac{1}{n} \sum_{x \in s} \mathbf{Z}(x) \mathbf{Z}(x)^\top \right\}^{-1} \frac{1}{n} \sum_{x \in s} \mathbf{Z}(x) Y(x)$$

By a **first-order Taylor expansion** (see Mandallaz p. 124-125 for derivation),

$$\hat{\beta}_s - \beta \approx \mathbf{A}_s^{-1} \frac{1}{n} \sum_{x \in s} R(x) \mathbf{Z}(x) \quad \text{where we define} \quad \mathbf{A}_s := \frac{1}{n} \sum_{x \in s} \mathbf{Z}(x) \mathbf{Z}(x)^\top$$

Design-based variance-covariance estimate for $\hat{\beta}_s$

$$\hat{\Sigma}_{\hat{\beta}_s} \approx \mathbf{A}_s^{-1} \left[\frac{1}{n^2} \sum_{x \in s} \hat{R}(x)^2 \mathbf{Z}(x) \mathbf{Z}(x)^\top \right] \mathbf{A}_s^{-1}$$

The classical regression estimator

Sampling design

Uniform iid sampling

Assisting model

$$\hat{Y}(x) := \mathbf{Z}(x)^t \hat{\beta}_s + R(x)$$

IMPORTANT property

$$\frac{1}{n} \sum_{x \in s} \hat{R}(x) = 0 \quad \text{if model includes intercept on } s$$

Point estimator

$$\hat{Y}_{reg} = \bar{\mathbf{Z}}^t \hat{\beta}_s \quad \text{where} \quad \bar{\mathbf{Z}} := \frac{1}{A_T} \int_T \mathbf{Z}(x) dx$$

Variance estimator

$$\hat{\mathbb{V}}[\hat{Y}_{reg}] = \bar{\mathbf{Z}}^t \hat{\Sigma}_{\hat{\beta}_s} \bar{\mathbf{Z}} \quad (\text{accounts for internal fit})$$

Note: $\bar{\mathbf{Z}}^t \hat{\Sigma}_{\hat{\beta}_s} \bar{\mathbf{Z}}$ is the matrix version of the identity $\mathbb{V}[c \cdot X] = c^2 \cdot \mathbb{V}[X]$

Classical regression estimator has good statistical properties...

Defined $g(x) = \bar{\mathbf{Z}}^t \mathbf{A}_s^{-1} \mathbf{Z}(x)$ and re-write as :

$$\hat{Y}_{reg} = \frac{1}{n} \sum_{x \in s} g(x) Y(x)$$

$$\hat{V}[\hat{Y}_{reg}] = \frac{1}{n^2} \sum_{x \in s} g(x)^2 \hat{R}(x)^2$$

Calibration properties of g-weights

$$\frac{1}{n} \sum_{x \in s} g(x) \mathbf{Z}(x) = \bar{\mathbf{Z}}$$

$$\frac{1}{n} \sum_{x \in s} g(x) = 1$$

The g-weighted sample exactly reproduces the known population means in the auxiliary vector.

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Great news if $\mathbf{Z}(x)$ and $Y(x)$ are well-correlated!

Post-stratification : classical regression estimation with categorical variables (i.e. ANOVA).

$$\hat{Y}_{reg} = \sum_{h=1}^H p_h \bar{Y}_{s,h} \quad \text{where} \quad p_h = \frac{A_h}{A_T}, \quad n_h := \# \text{ of plots in stratum } h.$$

External plug-in approach

$$\hat{\mathbb{V}}_{\text{ext}} = \frac{1}{n} \sum_h \frac{n_h - 1}{n - 1} \hat{\mathbb{V}}_h[\hat{R}(x)]$$

Disturbing features

- Strata weights based on sample, not population.
- Strata variance is proportional to n_h .

G-weight variance

$$\hat{\mathbb{V}}_g = \sum_h \frac{p_h^2}{n_h} \hat{\mathbb{V}}_h[\hat{R}(x)]$$

Better calibration properties

- Uses the **known census proportions** p_h .
- Stratum variance is proportional to $\frac{1}{n_h}$.

Special cases of the classical regression estimator

With **known wall-to-wall auxiliary information**, several important special cases arise:

One-phase estimation

- Intercept only
- No auxiliary information required

Regression estimation

- Continuous variables allowed
- Multiple linear regression w/ intercept

Post-stratification

- Categorical variables only
- ANOVA model
- Extremely common in NFIs

Regression within strata

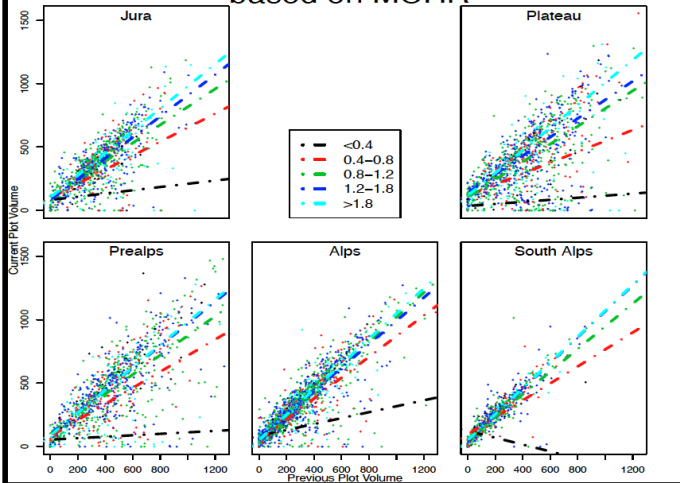
- Categorical + continuous variables
- Different intercepts by stratum
- Interactions allow different slopes

Don't choose a special case — choose the model to minimize residuals.

NOTE: We only need to implement the g-weight variance formula once!

Visual example of “regression in strata”

Proposed disturbance classes based on MCHR



The Swiss NFI uses permanent plots and had switched from periodic to annual panel design. In this context, previous volume measurement became a powerful predictor of current volume. Disturbance classes were identified using mean canopy height ratio (MCHR) between aerial campaigns.

Linear model with interactions :

- disturbance class
- region
- previous volume

Credit: Swiss Federal Institute for Forest, Snow and Landscape Research WSL

Today

- **Model-assisted estimation** improves precision by exploiting known totals/means of auxiliary information.
- Fundamentally **design-based**: the model shifts the estimation problem to the residuals.
- The model can be wrong — the estimator remains **(asymptotically) design-unbiased**.

Tomorrow

- **Multi-phase sampling**: auxiliary information available at **multiple sampling intensities**, rather than exhaustively wall-to-wall.
- Examine model-assisted techniques for **small-area estimation**.

BONUS: a taste of the classical survey sampling theory (Cordy, 1993)

- For fixed sample size n , denote the sample $s = \{x_1, x_2, \dots, x_n\} \subset T$.
- Under **uniform i.i.d.** sampling, every x_k has the same marginal density $f(x) = 1/A_T$.
- The **sampling intensity** (*inclusion prob.* from finite theory) is

$$\pi(x) = \sum_{k=1}^n f(x) = n f(x) = \frac{n}{A_T}, \quad x \in T$$

- The **second-order sampling intensity** (*inclusion prob.* of a pair being jointly sampled) is

$$\pi(x, x') = \sum_{k=1}^n \sum_{j \neq k} f(x) f(x') = n(n-1) f(x) f(x') = \frac{n(n-1)}{A_T^2}, \quad x \neq x'$$

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Note

General survey sampling allows $f_k(x)$ to vary by draw or be non-uniform (importance sampling, unequal probability designs). Under uniform i.i.d. sampling both intensities collapse to constants — they don't depend on x at all.

Horvitz–Thompson estimator and variance

General form

$$\hat{Y} = \sum_{x \in s} w(x) Y(x)$$

The **Horvitz–Thompson estimator** sets $w(x) = 1/\pi(x)$ and is design-unbiased whenever $\pi(x) > 0$ for all $x \in T$.

Horvitz–Thompson estimator and variance

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Horvitz–Thompson variance estimator

Unbiased under regularity conditions (most notably $\pi(x, x') > 0$):

$$\hat{\mathbb{V}}(\hat{Y}_\pi) = \sum_{x \in s} \left(\frac{Y(x)}{\pi(x)} \right)^2 (1 - \pi(x)) + \sum_{x \in s} \sum_{x' \neq x} Y(x) Y(x') \left(\frac{\pi(x, x') - \pi(x)\pi(x')}{\pi(x, x')\pi(x)\pi(x')} \right)$$

Several minutes later...

Plug in uniform i.i.d. intensities

With $\pi(x) = n/A_T$ and $\pi(x, x') = n(n-1)/A_T^2$, tedious algebra recovers exactly what we already had from Monte Carlo integration:

$$\hat{Y}_\pi = A_T \cdot \hat{Y}_{1p} = A_T \cdot \frac{1}{n} \sum_{x \in s} Y(x) \qquad \hat{\mathbb{V}}(\hat{Y}_\pi) = \hat{\mathbb{V}}[\hat{Y}_{1p}] = \frac{\hat{\mathbb{V}}[Y(x)]}{n}$$

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Why bother?

This is how most people construct their understanding for forest inventory sampling (although usually for finite population theory).

—

The **Monte Carlo approach** lets you more easily visualize, build intuition and derive a surprising number of valuable estimators.